

ESTIMATING SPAIN'S INCOME DISTRIBUTION FROM NEIGHBORHOOD-LEVEL DATA

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Abstract

This methodology note develops a transparent and simple framework to estimate aggregate income distributions in Spain exploiting the availability of granular inequality metrics. Specifically, I leverage administrative tax records and demographic data at the census tract-level to reconstruct local income distributions assuming log-normality. National, regional and municipal income distributions can be subsequently derived as population-weighted mixtures of these tract-level distributions. This approach leverages well-documented regularities in how incomes are distributed within demographically homogeneous groups, and the resulting estimates closely match observed distributional statistics in validation tests. I implement this framework in comparatuingreso.es, a publicly available web platform that enables Spanish households to calculate their relative position within the income distribution. While the focus is on Spain, this approach can be readily adapted to other countries with comparable data available.

1. Introduction

Understanding the income distribution is crucial for economic analysis and policy design, yet its measurement remains challenging. In this paper, I propose a transparent and simple method to estimate national-level income distributions when granular income and inequality measures are available. The approach builds on two key insights: first, that income distributions within demographically homogeneous areas tend to follow log-normal distributions; and second, that aggregate distributions can be effectively modeled as mixtures of these local distributions. Using detailed administrative data from over 36,000 Spanish census tracts, I demonstrate that local income distributions are well-approximated by log-normal distributions, with predicted statistics matching observed values within 5% on average. This method provides a flexible parametric framework for estimating income distributions when individual data is unavailable.

The rest of this note is organized as follows. Section 2 describes the data sources, their coverage, and scope. Section 3 outlines the methodology, including the modeling of local income distributions, their aggregation and the calculation of the relevant validation metrics. Section 4 presents the results. Finally, Section 5 concludes the note.

2. Data

The data used in this project is sourced from the Spanish Statistical Office (INE) Household Income Distribution Atlas (*Atlas de Distribución de Renta de los Hogares*, ADRH).¹ This dataset combines administrative tax data with population statistics to provide detailed information about the income distribution and related socioeconomic indicators at granular geographic levels in Spain.

Spain is administratively divided into several territorial levels that structure its governance and statistical reporting. At the highest level are the *Comunidades Autónomas* (autonomous communities), which are regions with significant legislative and executive powers granted by the Spanish Constitution. There are 17 autonomous communities and two autonomous cities (Ceuta and Melilla). These regions are further subdivided into *provincias* (provinces), the primary intermediate level of governance. Each province consists of *municipios* (municipalities), the fundamental local administrative units, which vary widely in size and population.

Below the municipal level, finer geographic divisions provide additional granularity. For statistical purposes, municipalities are divided into *distritos* (districts), which are then further subdivided into *secciones censales* (census tracts). Census tracts, encompassing areas with populations typically ranging from 1,000 to 2,500 residents, provide the finest level of geographic detail available in the ADRH.

Income data in the ADRH is derived from tax declarations submitted to the Spanish tax authorities, including the *Agencia Estatal de Administración Tributaria* (AEAT) and the Foral tax authorities.² These tax records provide detailed information on income from various sources, including wages,

¹Available online at https://www.ine.es/experimental/atlas/experimental_atlas.html/.

²The latter operate in regions with special fiscal regimes, such as the Basque Country and Navarre.

pensions, unemployment benefits, and other forms of revenue subject to the *Impuesto sobre la Renta de las Personas Físicas* (IRPF), Spain's personal income tax system. The dataset excludes non-resident income and focuses exclusively on individuals considered fiscal residents within the territory.

Demographic data in the ADRH is constructed from the *Fichero Precensal de Población* (FPC), a comprehensive register derived from the municipal register (*Padrón*) and other administrative sources. The FPC forms the basis of Spain's population census and ensures alignment between demographic and income data. The temporal reference for income data corresponds to the calendar year, while demographic data is anchored to the population as registered on January 1 of the following year. Individuals residing in collective establishments, such as nursing homes, hospitals, or military barracks, are excluded from the dataset.

Summary statistics. The sample consists of 37,072 census tracts covering all of Spain in 2023. The average tract has 1,315 residents, though size varies considerably (SD = 663). Average net income per equivalent adult is €22,036, about 48% higher than per capita income (€14,030). Within-tract inequality, measured by the Gini coefficient, averages 28.6 points across tracts. Missing rates are below 2% for most variables, and rise slightly above 5% for equivalized income and the Gini coefficient. Demographically, the average tract has a dependency ratio of 0.63, a mean age of 45.6 years, and 30.7% single-person households.

TABLE 1. Summary statistics

	(1)	(2)	(3)	(4)	(5)
	Min	Max	Mean	SD	% missing
<i>Income distribution</i>					
Net income per capita	5,522.00	36,918.00	15,025.25	4,431.91	1.83
Net income per equivalent adult	8,797.00	58,752.00	22,036.40	6,915.25	5.52
Gini	19.70	44.20	28.61	3.87	5.52
<i>Demographics</i>					
Population	2.00	12,327.00	1,315.16	678.37	1.60
Dependency ratio	0.00	4.00	0.63	0.18	1.60
Mean age	27.90	74.80	45.61	5.46	1.60
Single-person households (%)	0.00	100.00	30.70	9.58	1.60
No. of tracts	37,072				

Notes: all statistics are unweighted and calculated at the census-tract level. Spanish Statistical Office (INE) Household Income Distribution Atlas and author's calculations.

To improve coverage, I conduct two simple adjustments. First, I impute net income per equivalent adult within missing tracts using net income per capita and the average provincial ratio between these two measures, calculated using population weights from non-missing observations. This adjustment reduces missing rates in equivalised income per capita from 5.2% to 1.63%. Second,

I impute missing Gini coefficients (5.2% of tracts) using a machine learning approach. More details can be found in Section 3.

3. Methodology

3.1. Background

The study of income distributions has been a cornerstone of economic research. A long-standing empirical regularity in the literature is that income distributions tend to approximate a log-normal density (Aitchison and Brown, 1957). This stylized fact is supported by both theoretical and empirical foundations. At its core, log-normality arises from the multiplicative interaction of economic factors – such as human capital, local labor market conditions, and productivity – which determine individual incomes (Neal and Rosen, 2000). Gibrat’s Law of Proportionate Effect (Gibrat, 1931) formalizes the theoretical basis for log-normality through multiplicative random growth, showing that when the growth rate of a variable is independent of its initial size and the logarithm of the growth rate is independent and identically distributed over time with finite variance, the resulting distribution tends to converge to log-normality.

More concretely, the multiplicative processes through which log-normality arises operate most strongly within demographically homogeneous groups where individuals are subject to similar economic shocks (Weiss, 1972; Aitchison and Brown, 1957; Battistin, Blundell, and Lewbel, 2009). This methodology note relates to a larger body of research providing empirical support for this pattern, showing that a mixture of log-normals is particularly suitable for modeling income in fairly homogeneous sub-populations while at the same time accounting for between-group heterogeneity (Flachaire and Nuñez, 2007; Lubrano and Ndoye, 2016). This finding is particularly relevant at higher levels of aggregation, where finite mixtures of log-normals provide substantial improvements in fit compared to single log-normal specifications (Gardini, Fabrizi, and Trivisano, 2022).

The literature of neighborhood effects offers a complementary explanation for why such homogeneity in income-generating processes exists at local levels. In particular, neighborhood effects reinforce homogeneity within local populations through contextual influences and endogenous spillovers, which emerge from behavioral interactions within the neighborhood, such as peer influences or social norms (Manski, 1993; Durlauf, 1996). These mechanisms create reinforcing feedback loops that generate correlated income trajectories within neighborhoods – for instance, through shared information about job opportunities, similar human capital accumulation patterns, or common responses to local economic shocks. The resulting interaction structures generate strong within-neighborhood homogeneity while maintaining between-neighborhood heterogeneity (Durlauf, 2004). This pattern provides a rationale for estimating aggregate income distributions as mixtures of local-level distributions.

However, while providing a good approximation for much of the income distribution, the log-normal approximation exhibits systematic deviations in the tails. Since Pareto’s (1896) seminal work, research has shown that top incomes follow a power law rather than a log-normal decay

(Gabaix, 2016). Similarly, studies using detailed administrative tax data have documented that the upper tail of income distributions across countries and time periods are better described by a Pareto distribution (Atkinson, Piketty, and Saez, 2011). This departure from log-normality at high incomes is important for accurately measuring top income inequality, but modeling the precise behavior of the upper tail is beyond the scope of this note.

3.2. Empirical approach

I estimate the national income distribution by exploiting the granular structure of census tract-level data and the theoretical properties of income distributions. My approach leverages the fact that income distributions within small geographic units tend to follow log-normal distributions more closely than in larger areas, and approximates the aggregate distribution using a mixture of these local-level distributions.

Consider a census tract j with an observed *median* income $\tilde{m}_j$ and Gini coefficient G_j . Under the log-normality assumption, if income Y follows a log-normal distribution with parameters (ν_j, σ_j^2) , then

$$(1) \quad \ln(Y) \sim N(\nu_j, \sigma_j^2).$$

The mapping between these parameters and observed statistics is:

$$(2) \quad \tilde{m}_j = \exp(\nu_j),$$

$$(3) \quad G_j = 2\Phi\left(\frac{\sigma_j}{\sqrt{2}}\right) - 1,$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function. From the observed Gini coefficient, I recover σ_j as

$$(4) \quad \sigma_j = \sqrt{2} \Phi^{-1}\left(\frac{G_j + 1}{2}\right),$$

and from the median I identify $\nu_j = \log \tilde{m}_j$. When the tract median is unavailable but the (equalised) *mean* income μ_j is observed, I use the exact log-normal relationship

$$(5) \quad \mu_j = \exp\left(\nu_j + \frac{\sigma_j^2}{2}\right) \Rightarrow \nu_j = \log \mu_j - \frac{\sigma_j^2}{2}.$$

I then estimate the national distribution as a population-weighted mixture of these local log-normal distributions. For a given income level y , the density is

$$(6) \quad f(y) = \sum_{j=1}^J w_j f_j(y | \nu_j, \sigma_j^2),$$

where w_j is tract j 's population share and $f_j(\cdot | \nu_j, \sigma_j^2)$ is the log-normal density. Percentiles of the national distribution are obtained by solving

$$(7) \quad p = F(q_p) = \sum_{j=1}^J w_j \Phi\left(\frac{\ln(q_p) - \nu_j}{\sigma_j}\right),$$

where q_p is the p^{th} percentile and $F(\cdot)$ is the cumulative distribution function of the mixture. I use numerical root-finding methods to find the value of q_p that satisfies $F(q_p) - p = 0$ within a bounded interval.

Missing outcomes. For census tracts with missing Gini coefficients (approximately 5.5% of the sample), I estimate them using a machine learning approach. Specifically, I train an XGBoost model using the following demographic predictors: dependency ratio (ratio of population under 18 and over 65 to working-age population), mean age, percentage of single-person households, logged mean equivalised income, mean household size, population size, and province fixed effects. Model performance is validated using a 5-fold cross-validation procedure, and is trained using default hyperparameters without grid search or additional tuning. For comparison, I also estimate a baseline OLS model using the same set of predictors. Out-of-sample prediction errors under both approaches are shown in Table 2. Based on the cross-validation results, the XGBoost model demonstrates superior predictive performance compared to a baseline OLS model, as illustrated by a 6% lower RMSE.

TABLE 2. Performance comparison between OLS and XGBoost models

Model	RMSE	MAE	MAPE (%)
OLS	2.98	2.29	8.02
XGBoost (CV)	2.80	2.14	7.51
Relative improvement over OLS (%)	6.04%	6.55%	6.36%

Notes: This table compares the out-of-sample performance of two predictive models – OLS and XGBoost – on predicting missing Gini coefficients using demographic predictors. RMSE represents the root mean square error, MAE is the mean absolute error, and MAPE is the mean absolute percentage error. Relative improvements in each metric are calculated as the percentage reduction of the error metric when using XGBoost compared to OLS. Results are based on a 5-fold cross-validation procedure for the XGBoost model.

To validate the log-normality assumption at the tract level, I compare observed distributional statistics with their theoretical counterparts under log-normality. For each tract j , I recover (ν_j, σ_j) using the observed median and Gini coefficient. I then compute the following predicted statistics:

$$(8) \quad \widehat{\text{Mean}}_j = \exp\left(\nu_j + \frac{\sigma_j^2}{2}\right),$$

$$(9) \quad \widehat{\text{P80/P20}}_j = \exp\left(\sigma_j [\Phi^{-1}(0.8) - \Phi^{-1}(0.2)]\right).$$

To assess the fit, I estimate OLS regressions of the form

$$(10) \quad \hat{y}_j = \alpha + \beta y_j + \epsilon_j,$$

where $\hat{y}_j$ is the predicted value under log-normality and y_j is the observed value (either the tract mean income or the P80/P20 ratio).

3.3. Nowcasting to the current year

The ADRH is published with a lag of roughly two years, so the most recent available edition refers to income year 2023. To compare users against a contemporaneous distribution, I project tract-level incomes forward to the latest year for which aggregate income growth is observed (2024) using the Spanish Living Conditions Survey (*Encuesta de Condiciones de Vida*, ECV, the national component of EU-SILC). The ECV reports mean equivalised income (*renta media por unidad de consumo*) by autonomous community, with survey wave $t + 1$ referring to income earned in year t . For each autonomous community c , I compute the gross growth factor

$$(11) \quad g_c = \frac{\bar{y}_{c,2024}^{\text{ECV}}}{\bar{y}_{c,2023}^{\text{ECV}}},$$

and scale the income of every tract $j \in c$ by g_c . Because multiplying a log-normal variate by a constant shifts its location parameter, $v_j \mapsto v_j + \ln g_c$, while leaving the shape parameter σ_j unchanged, this operation preserves within-tract inequality — and hence the tract Gini coefficient — and updates only the level of the distribution. The upscaling factor is applied identically to the income of every census tract and municipality within an autonomous community, so that the entire small-area distribution shifts up while its shape is preserved.

To ensure that the implied national growth matches the aggregate ECV figure exactly, I rescale the community-level factors by a common constant,

$$(12) \quad \tilde{g}_c = k g_c, \quad k = \frac{g_{\text{ES}}}{\sum_c s_c g_c},$$

where g_{ES} is the ECV national growth factor and s_c is the share of autonomous community c in aggregate equivalised income, computed from the ADRH. Since the national mean of the mixture grows by the income-weighted average of the community factors, this construction guarantees $\sum_c s_c \tilde{g}_c = g_{\text{ES}}$: the national mean is anchored to the ECV national figure while relative differences in regional income growth are retained. In the 2023–2024 application, national mean equivalised income grew by 5.4%, with (calibrated) community-level factors ranging from -1.1% (Melilla) to $+8.3\%$ (Cantabria). This adjustment rests on two assumptions: that the year-on-year *growth* of survey-based equivalised income is a good proxy for growth in the administrative (tax-based) measure within each region — levels differ across the two sources, but growth rates are considerably more comparable — and that within-area inequality is stable over the one-year horizon.

4. Results

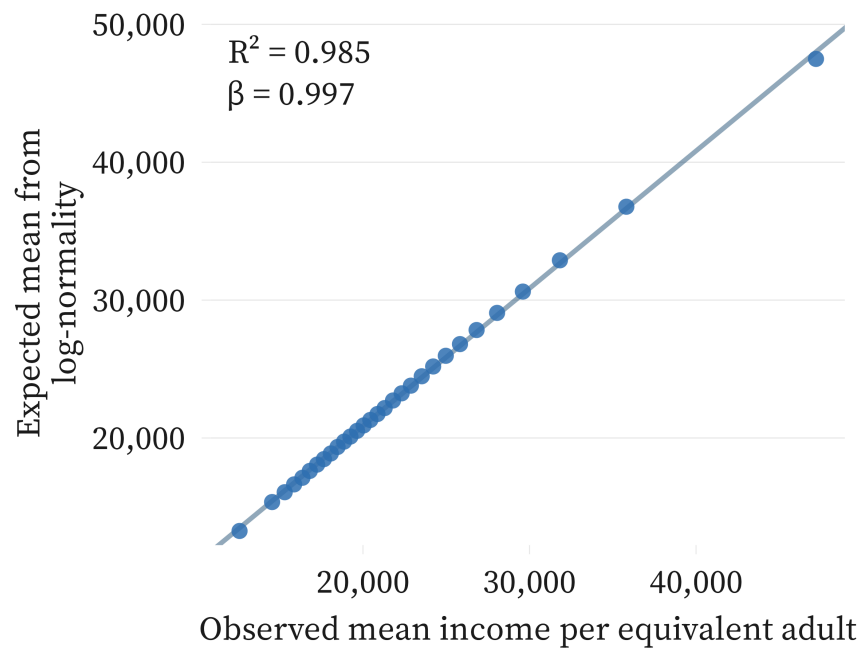
Validation. I begin by validating the log-normality assumption at the tract level. Figure 1 examines the fit for mean income, comparing observed values with those predicted under log-normality for each census tract, where observations are grouped into equal-sized bins using binscatter methods.³ The relationship is remarkably strong ($\beta = 0.99$, $R^2 = 0.99$), with predicted values differing from observed ones by just 4.6% on average. The near-unit slope coefficient and R^2 indicate that log-normality captures the central tendency of the income distribution particularly well.

Figure 2 shows the relationship for the P80/P20 ratio. The regression yields a slope coefficient of 0.77 ($R^2 = 0.82$), with predicted values deviating from observed ones by 4.8% on average. Consistent with the fact that real income distributions tend to have heavier tails, the coefficient below unity

³See Cattaneo et al. (2024).

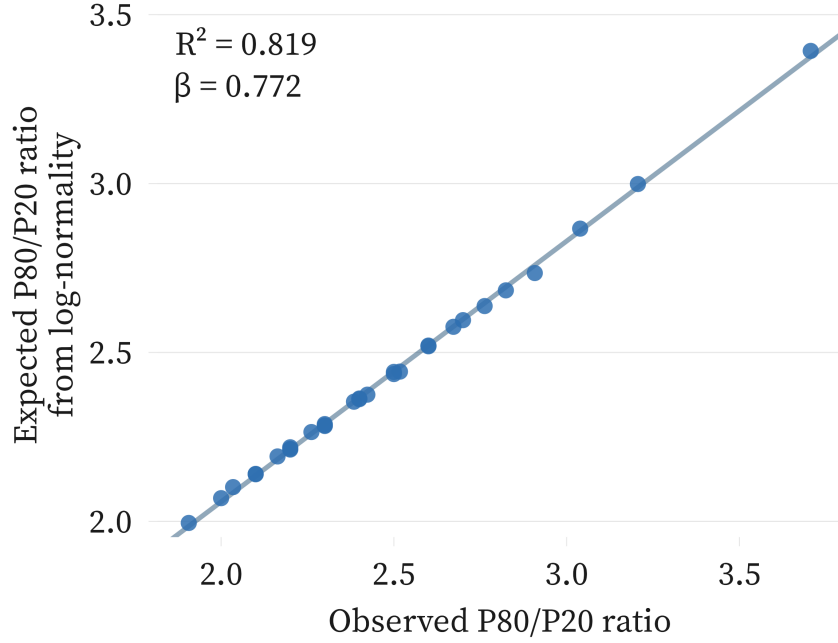
suggests that the log-normal distribution slightly underestimates inequality.

FIGURE 1. Validation of log-normality: mean income



Notes: This figure shows a binned scatter plot comparing observed mean income with predicted values under the log-normal assumption for each census tract. Source: Spanish Statistical Office and author's calculations.

FIGURE 2. Validation of log-normality: P80/P20 ratio



Notes: This figure shows a binned scatter plot comparing observed P80/P20 ratios with predicted values under the log-normal assumption for each census tract. Source: Spanish Statistical Office and author's calculations.

Decomposing income inequality. I examine the spatial structure of income inequality in Spain through a hierarchical variance decomposition exercise. Specifically, I decompose the total variance of log income within each autonomous community c into components corresponding to different geographic levels as follows:

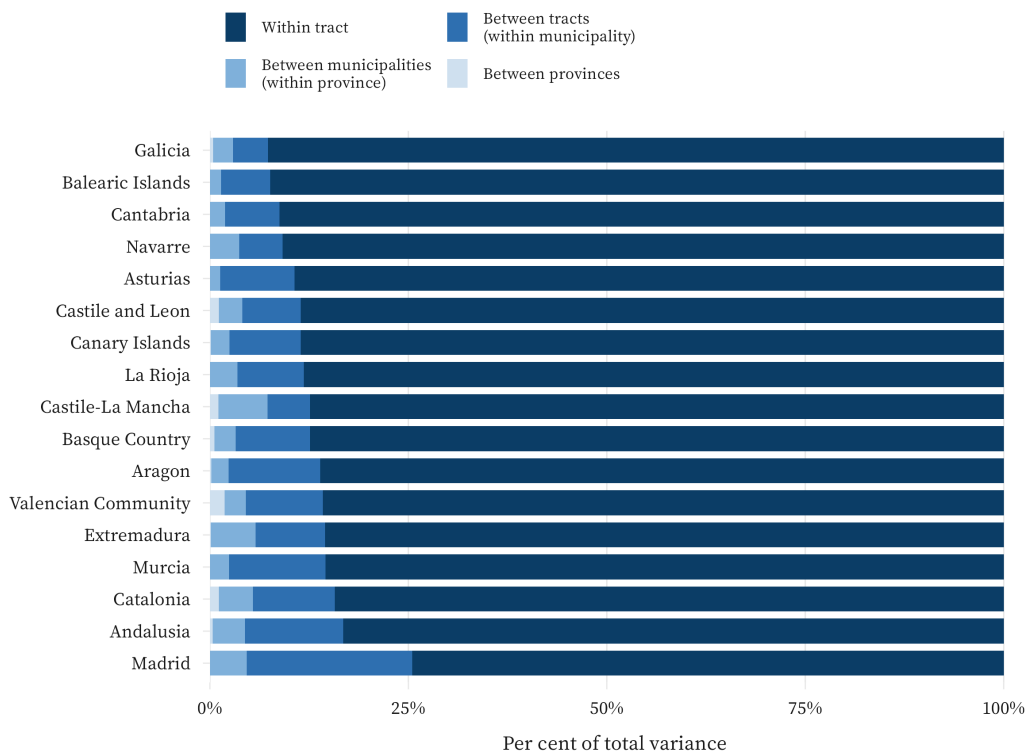
$$(13) \quad \sigma_c^2 = \underbrace{\sum_j w_j \sigma_j^2}_{\text{Within-tract}} + \underbrace{\sum_{k \in c} \sum_{j \in k} w_j (\mu_j - \mu_k)^2}_{\text{Between-tract}} + \underbrace{\sum_{p \in c} \sum_{k \in p} w_k (\mu_k - \mu_p)^2}_{\text{Between-municipality}} + \underbrace{\sum_{p \in c} w_p (\mu_p - \mu_c)^2}_{\text{Between-province}}$$

where σ_j^2 is the variance of log income within each census tract j , μ_j is the mean log income in tract j , μ_k is the mean log income in municipality k , μ_p is the mean log income in province p , and μ_c is the cross-tract mean log income within autonomous community c . Population weights for tracts, municipalities, and provinces, are denoted by w_j , w_k , w_p , respectively. The decomposition separates total income variation into four components: within-tract, between-tract (within municipalities), between-municipality (within provinces), and between-province variation.

Figure 3 presents the results of this decomposition by autonomous community. The analysis reveals that within-tract variation accounts by far for the largest share of income inequality, representing around 80% of total variance on average. This finding provides additional support for using

tract-level distributional estimates as the fundamental building blocks for the aggregate distribution.⁴ A further 11% is on average explained by between-tract, within-municipality differences. Income differences between municipalities and provinces contribute the remaining 9%.⁵

FIGURE 3. Variance decomposition



Notes: This figure shows the decomposition described in Equation 13 whereby total variance in log income is broken down into four components: within-tract, between-tract (within municipality), between-municipality (within province), and between-province variation. Source: Spanish Statistical Office and author's calculations.

5. Conclusion

This methodology note presents a simple framework for estimating income distributions in Spain, leveraging granular inequality metrics and administrative data at the census-tract level. Specifically, I model local income distributions as log-normal and derive national and regional distributions as population-weighted mixtures of tract-level estimates. Validation results show a high degree of accuracy, with predicted metrics closely matching observed distributional statistics across census

⁴Notice that this does not contradict the earlier discussion on homogeneity in income-generating *processes* within neighborhoods. The latter refers to the shared mechanisms (e.g., proportional shocks) influencing income trajectories within neighborhoods, but does not imply homogeneity in outcomes. Individual incomes can still vary significantly due to structural or stochastic factors.

⁵Extending the decomposition analysis at the national level shows that between-community income disparities only account for 5% of total variance.

tracts. Results from a hierarchical variance decomposition exercise show that within-tract variation explains the vast majority of income inequality in Spain.

Looking ahead, the methodology offers opportunities for extension, such as explicitly addressing the deviations from log-normality at the tails of the distribution or adapting the framework for use in other countries with comparable data available.

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